Test 1

No.1 Number of atoms in 558.5 g Fe (at.wt.55.85) is:

(a) Twice that in 60 g carbon
(b) 6.023×10^{22} (c) Half in 8 g He
(c) 558.5×10^{23}

No.2 Which of the following changes with increase in temperature:

(a) Molality
(b) Weight fraction of solute
(c) Fraction of solute present in water
(d) Mole faraction

No.3 In a compound C, H, N atoms are present in 9 : 1 : 35 by weight. Molecular weight of compound is 108. Its molecular formula is:

(a) $C_2H_6N_2$ (b) C_3H_4N
(c) $C_6H_8N_2$ (d) $C_9H_{12}N_3$

No.4 What volume of hydrogen gas at 273K and l atm pressure will be consumed in obtaining 21.6g elemental boron (at. masss = 10.8) from the reduction of boron trichloride with hydrogen:

(a) 44.8 L (b) 22.4 L

(c) 89.6 L (d) 67.2 L

No.5 25 mL of a solution of barium hydroxide on titration with 0.1 molar solution of hydrochloric acid gave a tltre value of 35 mL. The molarity of barium hydroxide is:

(a) 0.28 (b) 0.35

(c) 0.07 (d) 0.14

No.6 6.02×10^{20} molecules of urea are present in 100 mL solution. The concentration of urea solution is:

(a) 0.1 M (b) 0.01 M

(b) 0.02 M (d) 0.001 M

No.7 To neutralize completely 20 mL of 0.1M aqueous solution of phosphorus(H_3PO_3) acid the volume of 0.1 M aqueous KOH solution required is:

(a) 60 mL (b) 20 mL

(c) 40 mL (d) 10 mL

No.8. Two solutions of a substance (non – electrolyte) are mixed in following manner 480 mL of 1.5 M of first solution with 520 mL of 1.2 M of second solution. The molarity of final solution is:

(a) 1.20 M (b) 1.50 M

(c) 1.344 M (d) 2.70 M

No.9 If 1/6, in place of 1/12, mass of carbon atom is taken to be the relative atomic mass unit the mass of one mole of a substance will:

(a) Decrease twice (b) Increase two folds

(c) Remains unchanges

(d) Be a function of the molecular mass of element

No.10 How many moles of magnesium phosphate $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms:

(a) 0.02 (b) 3.125×10^{-2}

(c) 1.25×10^{-2} (d) 2.5×10^{-2}

No. 11 Density of 2.05 M solution of acetic acid in water is 1.02 g/mL. The molality of same solution is:

(a) 1.14 mol kg^{-1} (b) 3.28 mol kg^{-1}

(c) 2.28 mol kg^{-1} (d) 0.44 mol kg^{-1}

No. 12 The normality of 0.3 M phosphorous acid H_3PO_3 is:

(a) 0.1 (b) 0.9

(c) 0.3 (d) 0.6

No.13 An aqueous solution of 6.3 g oxalic acid dihydrate is made upto 250 mL. The volume of 0.1 N NaOH required to completely neutralize 10mL of this solution is:

(a) 40 mL (b) 20 mL

(c) 10 mL (d) 4 mL

No.14 How many moles of electron weigh one kilogram:

(a) 6.023×10^{23} (b) $\frac{1}{9.108} \times 10^{23}$

(c) $\frac{6.023 \times 10^{54}}{9.108}$ (d) $\frac{1}{9.108 \times 6.023} \times 10^{8}$

No.15 Which has the maximum number of atoms:

(a) 24 g C(12) (b) 56 g Fe (56)

(c) 27 g Al(27) (d) 108 g Ag (108)

Alpha Classes OMR

Mole Concept

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D

Alpha Classes OMR

Mole Concept-Answers

1	**A**	B	C	D
2	A	B	**C**	D
3	A	B	**C**	D
4	A	B	C	**D**
5	A	B	**C**	D
6	A	**B**	C	D
7	A	B	**C**	D
8	A	B	**C**	D
9	A	B	**C**	D
10	A	**B**	C	D
11	A	B	**C**	D
12	A	B	C	**D**
13	**A**	B	C	D
14	A	B	C	**D**
15	**A**	B	C	D

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